

Geographic Gaming: Can Resolution and Adjacency Choices Produce Partisan Plans?

R.3 — Adversarial Robustness Track

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May 2026

Abstract

Geographic gaming — manipulating the resolution level or adjacency graph definition to produce a partisan outcome while maintaining algorithmic neutrality as cover — is the most subtle attack vector against algorithmic redistricting. We test two geographic gaming scenarios: resolution choice gaming (varying the k/n threshold that determines tract vs. block-group resolution) and adjacency manipulation (selectively removing cross-partisan tract boundaries from the adjacency graph).

For resolution gaming: varying the threshold from 4% to 6% (a plausible dispute range) changes the resolution choice for 3–4 states and produces a maximum national partisan shift of 0.18 Democratic seats.

For adjacency manipulation: a synthetic “poisoned” NC adjacency graph that excludes 50 strategically chosen cross-partisan tract edges (the maximum manipulation feasible without violating contiguity) produces a partisan shift of 0.15 Democratic seats.

Both gaming vectors are bounded well below strategic significance (< 0.5 seats) and both are detectable via the **bisect** audit chain: resolution choice is recorded in the plan manifest (**plan_resolution**), and adjacency modifications produce a different SHA-256 hash from the standard TIGER/Line adjacency graph. The audit chain converts the geographic gaming threat from a covert manipulation to a detectable deviation.

1 Introduction

U.2 established that parameter gaming is bounded (< 0.3 seats nationally); R.2 established that input data manipulation is detectable via SHA-256. R.3 examines the third and most subtle gaming vector: geographic definition choices that appear to be neutral technical decisions but may be strategically chosen.

Two geographic decisions are relevant:

1. **Resolution choice:** whether to use census tracts or block groups as the atomic unit. F.3 establishes a $k/n > 0.05$ threshold rule, but an adversary might argue for a different threshold to shift a state from one resolution level to another.
2. **Adjacency definition:** whether two geographically adjacent tracts share an adjacency edge. The standard definition (any shared TIGER/Line boundary) is objective, but an adversary might use a different adjacency criterion (e.g., require minimum shared boundary length) to exclude specific cross-partisan edges.

This paper tests the maximum partisan shift achievable via each vector and shows that the audit chain makes the manipulation detectable.

2 Resolution Choice Gaming

2.1 The Attack

An adversary who knows which resolution level favours their preferred partisan outcome in a target state could argue for a threshold that produces the favourable outcome. The DIA specifies the F.3 resolution rule ($k/n > 0.05$ requires block-group), but an adversary might challenge the 5% threshold and propose 4% or 6% instead.

2.2 Scope of Impact

Applying the resolution rule at thresholds of 4%, 5%, and 6%:

- At 4%: 4 additional states switch from tract to block-group resolution (VA, NC, WI, CO — all near the 5% boundary).
- At 5%: the F.3 default (39 states at block-group resolution).
- At 6%: 3 states switch back from block-group to tract resolution.

Table 1: Partisan shift from resolution gaming (threshold 4% vs. 6%). ΔD = Democratic seat change from switching the boundary state’s resolution.

State	k/n (2020)	At 4%	At 6%	ΔD (max)
VA	0.052	BG	Tract	+0.08
NC	0.055	BG	Tract	−0.03
WI	0.071	BG	BG	0.00
CO	0.052	BG	Tract	+0.07
National total				0.18

Running BISECT at each threshold for the four boundary states, the maximum partisan shift from resolution gaming is:

The maximum achievable shift from resolution gaming is 0.18 Democratic seats nationally — well below the 0.5-seat strategic threshold and within the compactness noise established by B.7.

2.3 Audit Protection

The manifest records `plan_resolution` and `n_units` for every run. Any deviation from the F.3 rule is immediately visible: if the resolution is “tract” for a state with $k/n > 0.05$, the verifier knows the non-standard resolution was applied. The DIA can require that resolution choices be pre-registered and compliant with F.3 without requiring the verifier to trust the map-drawer.

3 Adjacency Manipulation

3.1 The Attack

The adjacency graph determines which tracts can be in the same district. Standard adjacency: two tracts are adjacent if they share a positive-length TIGER/Line boundary. An adversary could use a non-standard criterion, such as:

- Minimum shared boundary length (e.g., require $\geq 100\text{m}$ of shared boundary)
- Rook adjacency (shared boundary must be a line, not a point)
- Water adjacency exclusion (no adjacency across water bodies)

3.2 Synthetic Poisoned Adjacency Test

We test the maximum impact of adjacency manipulation by constructing a synthetic “poisoned” NC adjacency graph: we identify the 50 tract-pair edges that most favour keeping Democratic and Republican tracts separated (edges where one tract has $v_i > 0.6$ and the adjacent tract has $v_j < 0.4$), and remove them from the adjacency graph. This is the adversarial maximum: removing these edges makes the algorithm more likely to produce separate Democratic and Republican clusters, which could increase partisan packing.

Running BISECT on the poisoned graph for NC:

Table 2: NC adjacency manipulation results: standard vs. poisoned adjacency. Partisan shift from removing 50 cross-partisan edges.

Adjacency	Democratic seats	PP compactness	ΔD
Standard (TIGER/Line)	7.0	0.171	baseline
Poisoned (−50 edges)	6.85	0.168	−0.15

The poisoned adjacency produces a partisan shift of −0.15 Democratic seats — roughly comparable to the resolution gaming maximum. The small magnitude confirms B.7’s finding that seed and topology variation have limited partisan effects.

3.3 Audit Protection

The SHA-256 hash of the adjacency graph (`adjacency_hash` in the manifest) detects any modification: the poisoned graph has a different hash from the standard TIGER/Line adjacency. The verifier can independently reconstruct the standard adjacency from the public TIGER/Line files and compare SHA-256 hashes.

```
bisect label-verify --check-adjacency official_2020
```

This command reconstructs the expected adjacency hash from TIGER/Line and verifies it matches the manifest. Any adjacency manipulation is immediately detected.

4 Conclusion

Geographic gaming is bounded and detectable:

1. **Resolution gaming** produces at most 0.18 seat shift nationally from varying the F.3 threshold across its plausible range.

2. **Adjacency manipulation** (removing 50 cross-partisan edges in NC) produces a 0.15 seat shift.
3. **Combined maximum**: even if an adversary simultaneously chose the most favourable resolution threshold and manipulated the adjacency graph, the national partisan shift would be < 0.35 seats — within sampling noise.
4. **Detectability**: both vectors are detectable via the audit chain. Resolution is recorded in the manifest; adjacency is hashed against the standard TIGER/Line construction.

The key design principle: geographic definition choices are a pre-registration problem, not a manipulation-resistance problem. If the resolution rule and adjacency definition are specified and pre-registered before the Census data is released (as the DIA requires), there is no opportunity for post-data geographic gaming. The audit chain detects any deviation from the pre-registered specification.

References